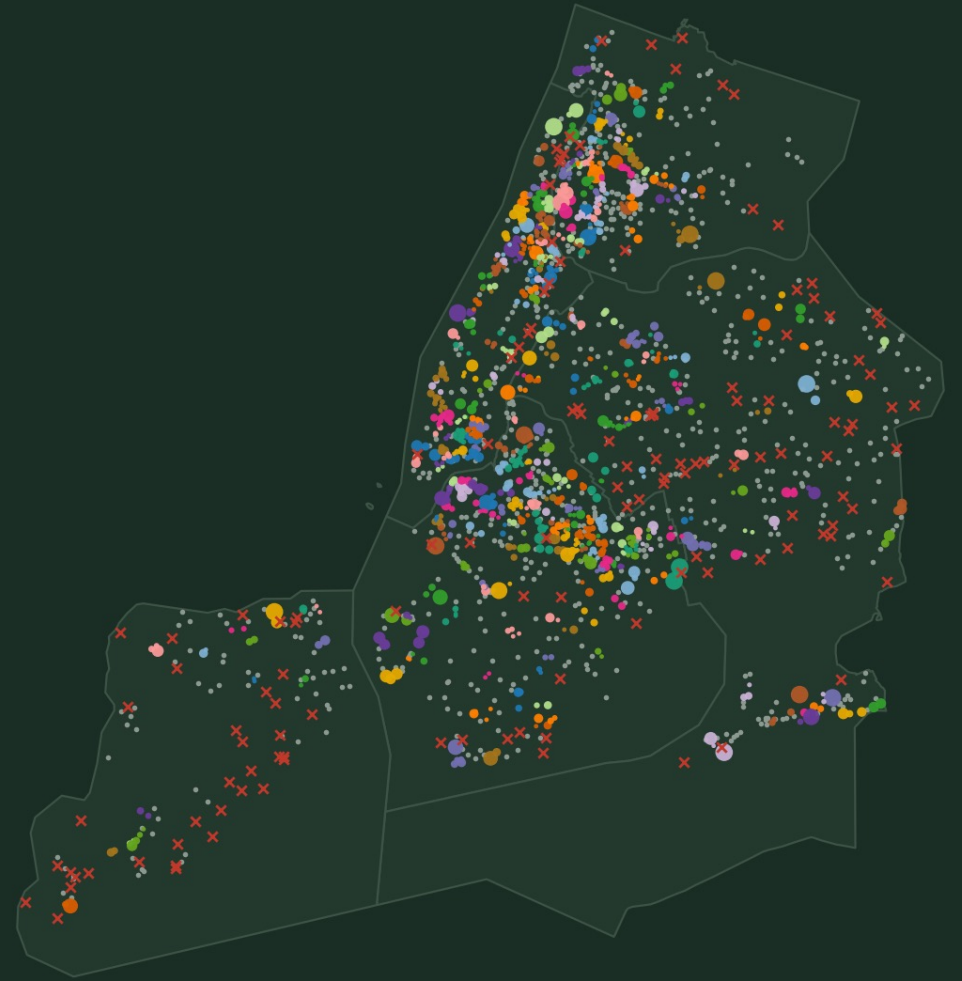


Configurational Park Networks in New York City

Which parks are linked by the street configuration — few-turn routes, shared corridors, shared fields — rather than by mere proximity?

2,057 park features · OSM pedestrian network · Cityseer 5.8 angular analysis
October 2026



Not proximity: parks related through street configuration

Question Which small & medium parks form configurational groups, and which remain isolated?

Principle Distance is only a search bound; the edge itself is angular / syntactic.

Three definitions N1 few-turn routes · N2 shared corridor · N3 shared few-turn field.

Robustness Every definition is a grid of parameter settings; groups and isolation are what persists across them.

Walkable Every pair of parks in a group is within a 10-minute walk (800 m).

N1

Angular few-turn

total turning on the simplest route $\leq \theta$

N2

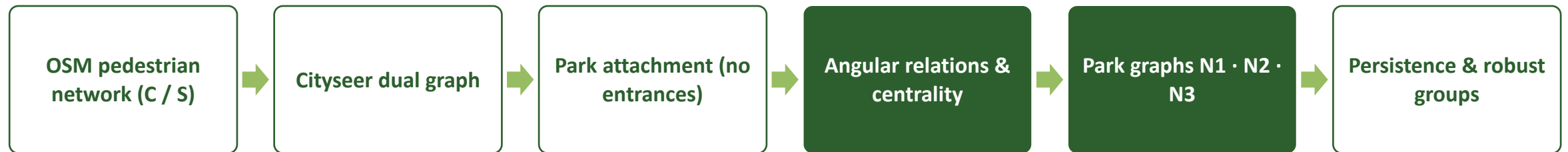
Shared corridor

both parks on the same high-NACH stroke

N3

Field overlap

few-turn catchments overlap more than distance predicts



Pipeline implemented as STEP notebooks 02–13 (Cityseer 5.8, OSMnx, GeoPandas, NetworkX).

Data: parks and two pedestrian network representations

1,902

parks analysed (of 2,061): flagship parks, non-park parcels and parks > 100 acres excluded

25 → 180

long linear parks (parkways, malls, esplanades) split into ~400 m pieces

496k

network segments in C (main); 1.02M in S with sidewalks

S · with sidewalks & crossings
2,054 segments



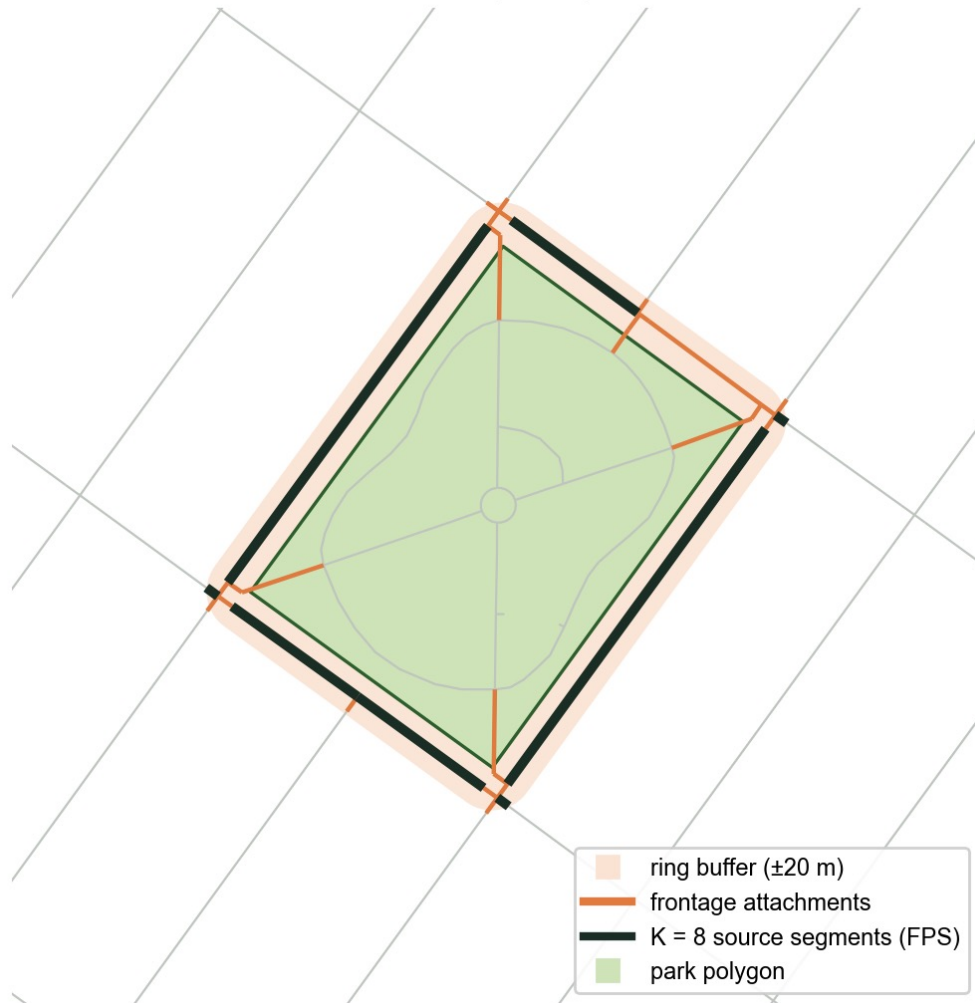
C · centrelines + pedestrian paths (main)
876 segments



S draws sidewalks and crossings separately: parks facing each other across a street are 180° apart. C (street centrelines + pedestrian-only paths) is the Space-Syntax-like segment map used for the main analysis.

Attaching parks to the network without entrance data

Maria Hernandez Park (6.9 ac): 36 attachments



1

Ring buffer

Segments within ± 20 m of the park boundary (± 10 m on S): the first ring of streets around the park.

2

Frontage rule

Keep a segment only if $\geq 50\%$ of it, or $\geq \min(2.5 \cdot d, 25\% \text{ of perimeter})$, lies in the ring — streets that merely clip a corner are dropped.

3

Size control

Up to $K = 8$ source segments per park by farthest-point sampling along the boundary; robustness measured as a share, not a count.

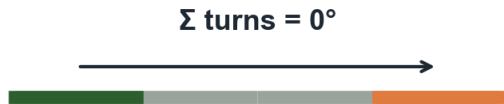
4

Unresolved $\neq$ isolated

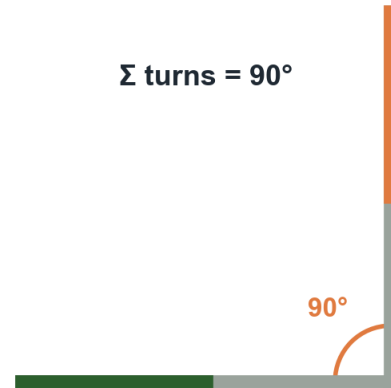
Parks attached only to tiny OSM fragments are flagged unresolved rather than isolated.

Angular cost: how many degrees does the simplest route turn?

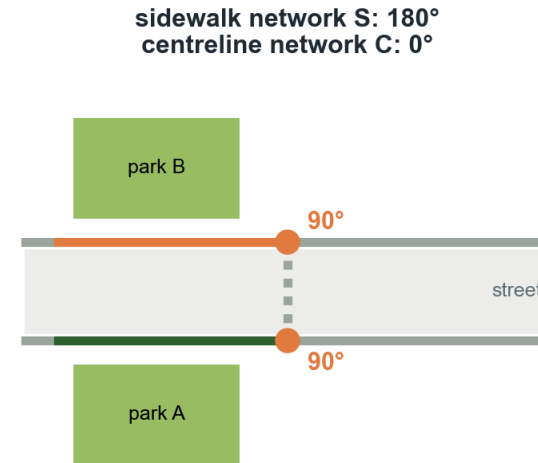
(a) Straight continuation



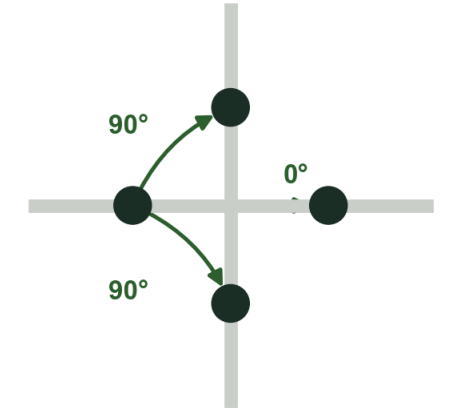
(b) One turn



(c) Crossing the street



(d) Cityseer dual graph



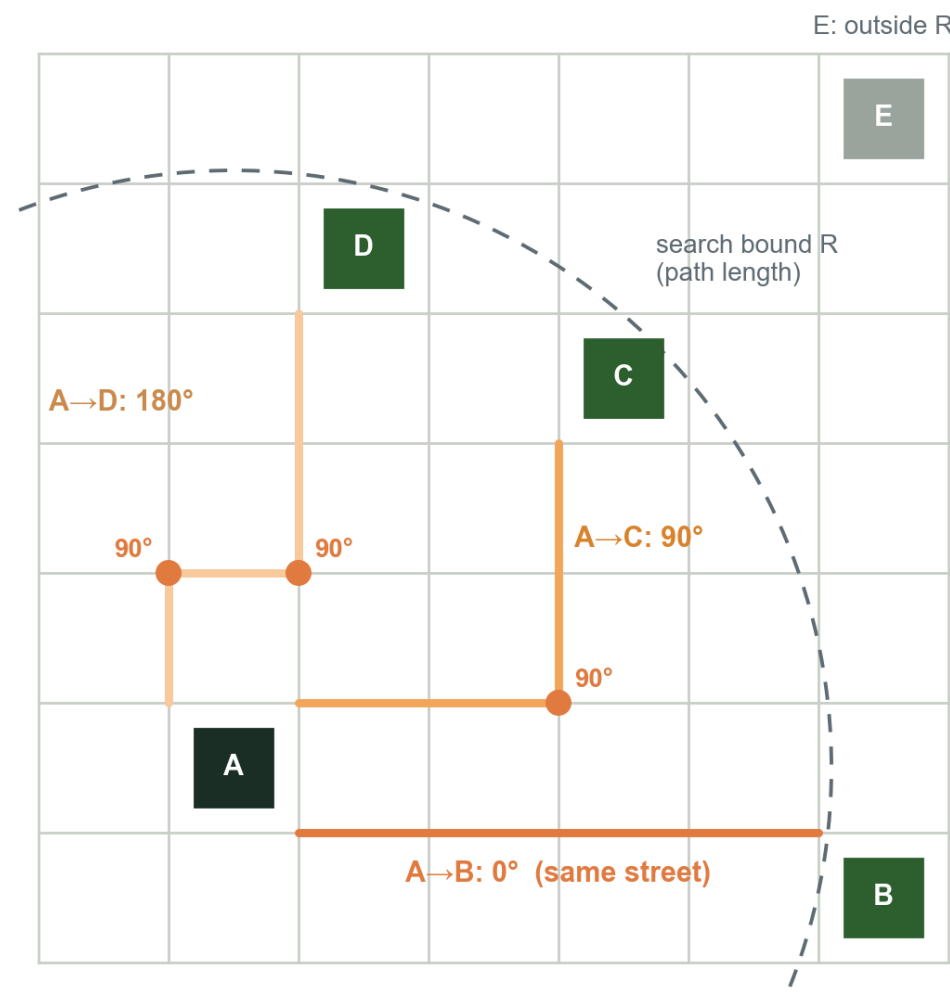
segments = nodes · turns = edge costs

Simplest path Cityseer routes on the dual graph (street segments = nodes) and minimises the cumulative change of direction, in degrees, including bends inside segments.

Metric bound Each search runs up to 3.2 km of walking; $R \in \{800, 1600, 3200\}$ m filters the path length afterwards — distance bounds the search, it does not define the edge.

N1 · Angular few-turn network

Two parks are linked if the simplest route between them turns little in total



$A_{ij}(R) = \min \text{ over source } u \in B_i, \text{ target } v \in B_j \text{ of angular cost}(u \rightarrow v), \text{ path length} \leq R$

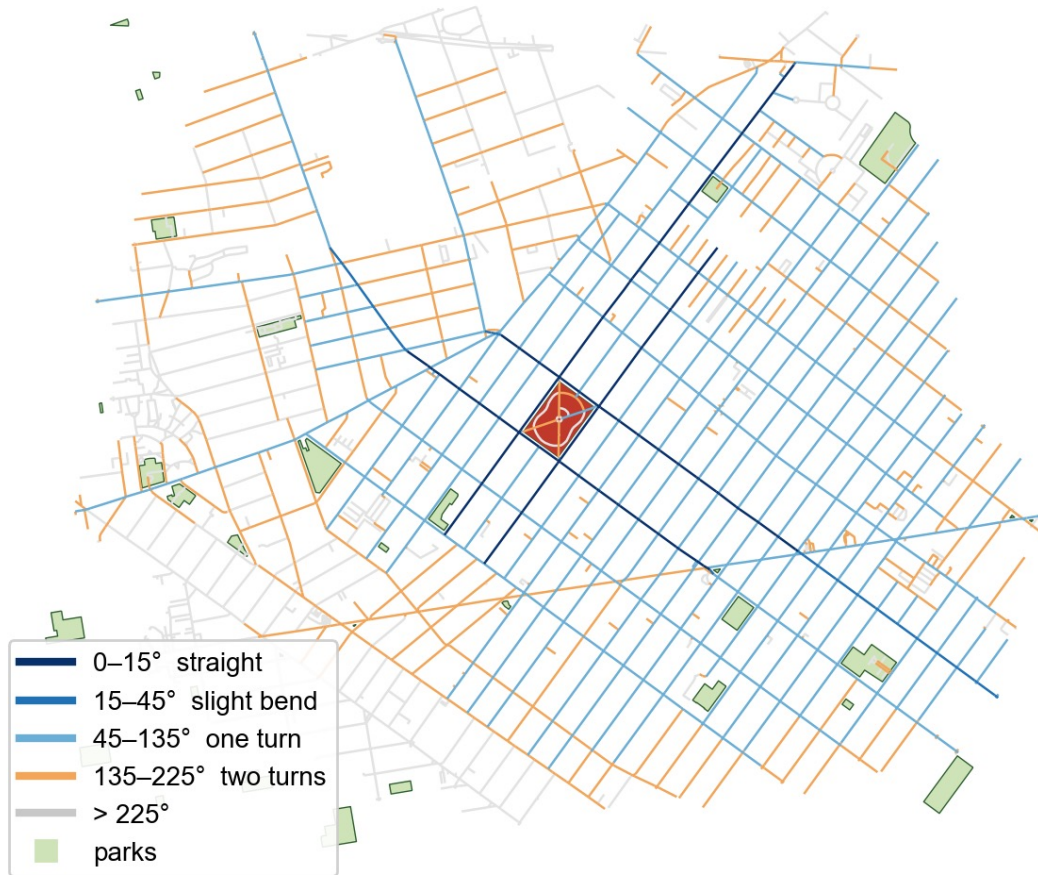
edge $\Leftrightarrow A_{ij} \leq \theta$

θ	reading	edges from A
15°	same straight street	B
45°	nearly straight	B
90°	one right-angle turn (strict)	B, C
180°	two turns	B, C, D

- 24 settings: $\theta \in \{15, 45, 90, 180\}^\circ \times R \in \{800, 1600, 3200\} \text{ m} \times \{\text{best pair}, \geq 2 \text{ robust pairs}\}$
- Trivial co-frontage (shared street segment, touching polygons) and pieces of the same park are excluded

N1 · How it is computed

Angular field from Maria Hernandez Park (simplest paths ≤ 1.6 km)



1

Angular trees

One simplest-path search per source segment (≤ 3.2 km) on borough networks with 3.2 km buffers

2

Park pairs

Minimum angle to every attachment of every other park $\rightarrow$ 212,137 directed park pairs

3

Threshold grid

Apply θ and R ; build 24 park graphs; classify isolated / dyad / chain / 2-core

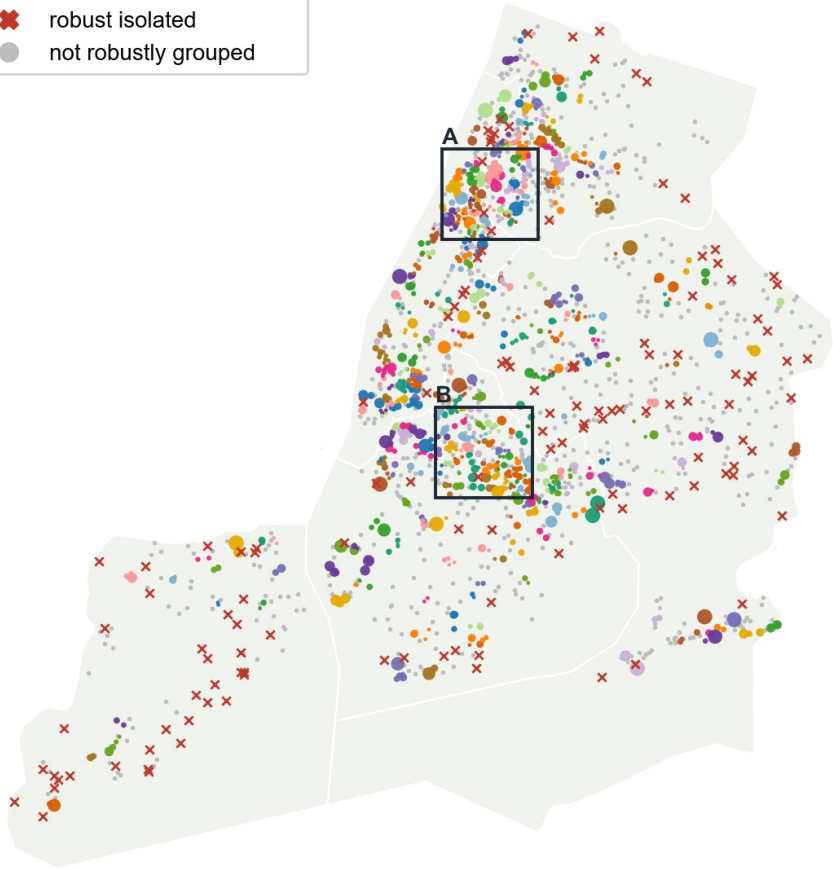
4

Groups

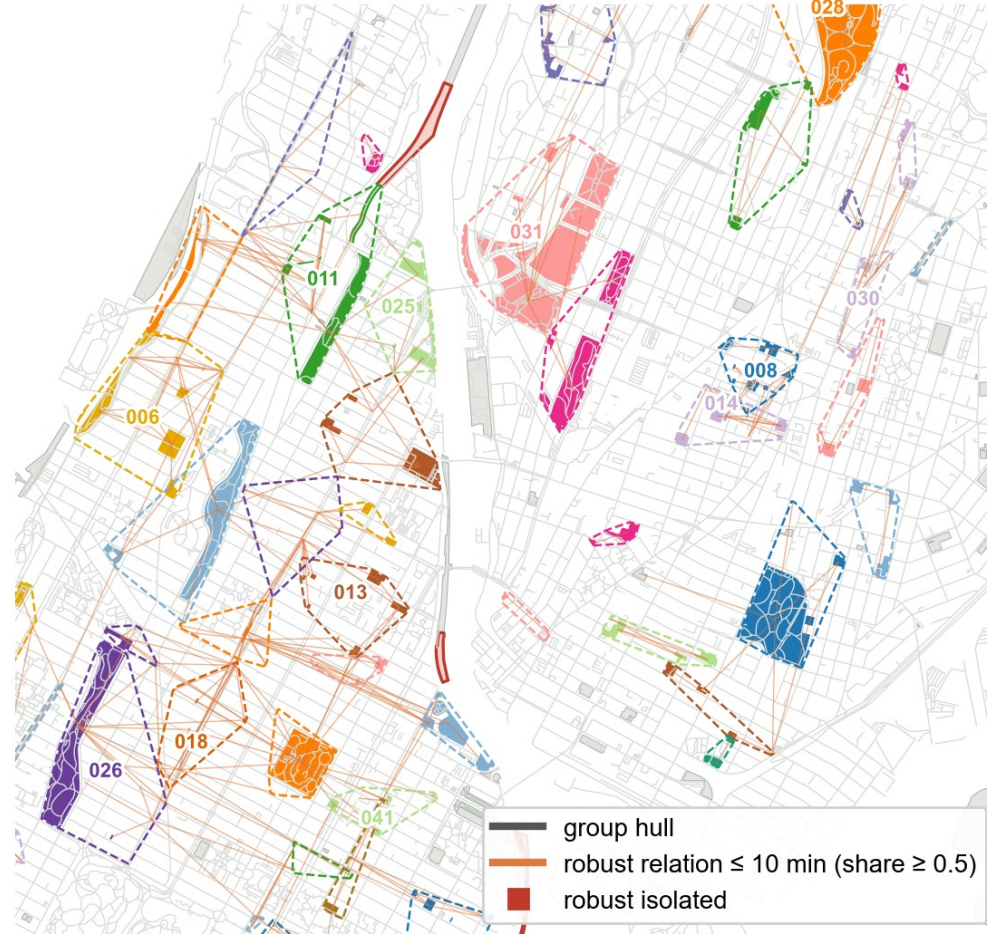
Communities on the 9 non-percolating settings (weights $\exp(-A / 45^\circ)$); consensus; split to a 10-min walking diameter

N1 · Groups of parks linked by few-turn routes

- grouped (colour = group)
- ✕ robust isolated
- not robustly grouped



A · Harlem / Upper Manhattan – South Bronx ($\gamma = 2$)



340

walkable groups ($\gamma = 2$, every pair ≤ 10 min)

1,178

parks in a group

391 m

median walking span within a group

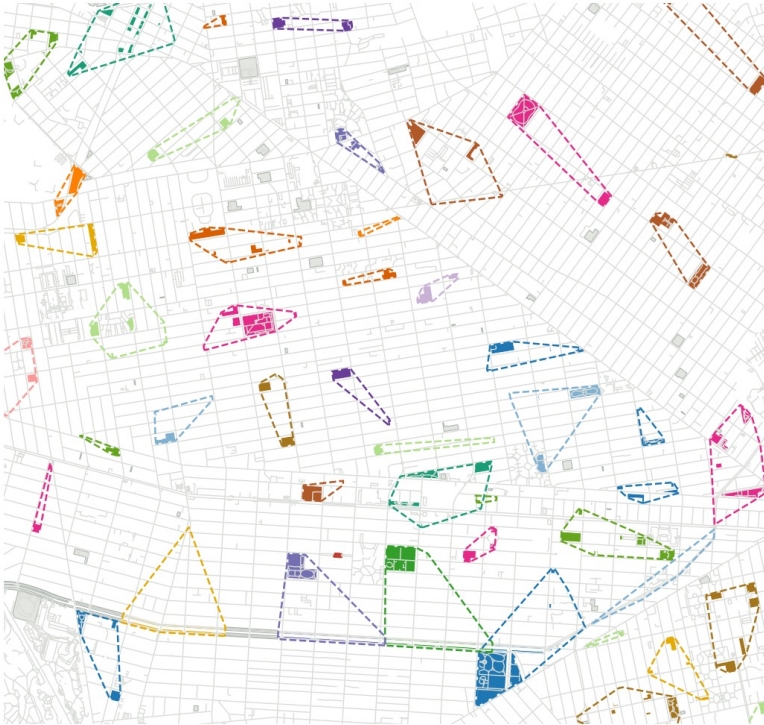
132

robust isolated ($\geq 80\%$ of settings)

N1 · Three scales of grouping

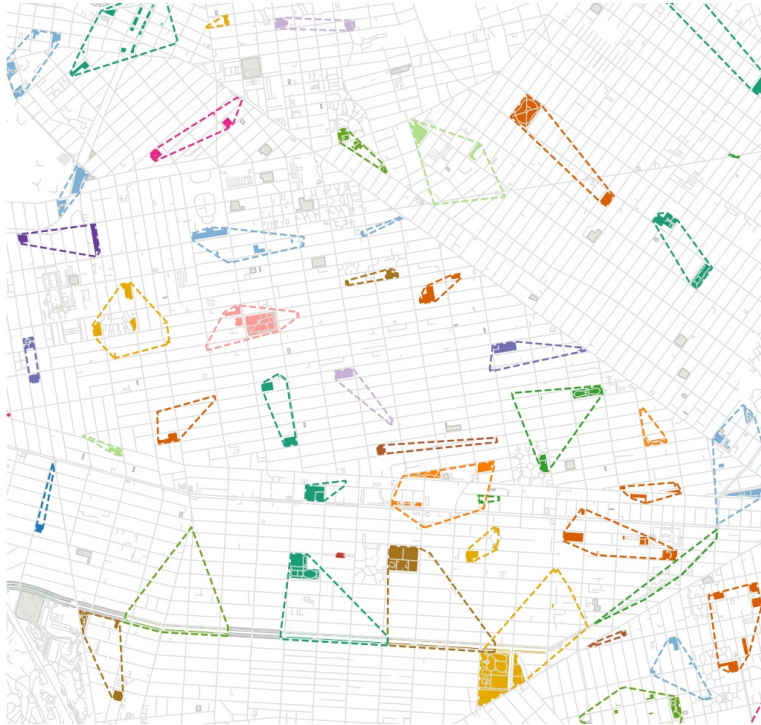
Louvain resolution γ : 1 = neighbourhood scale, 4 = block scale (zoom B)

$\gamma = 1$: 336 groups · max 15 parks · median extent 0.49 km



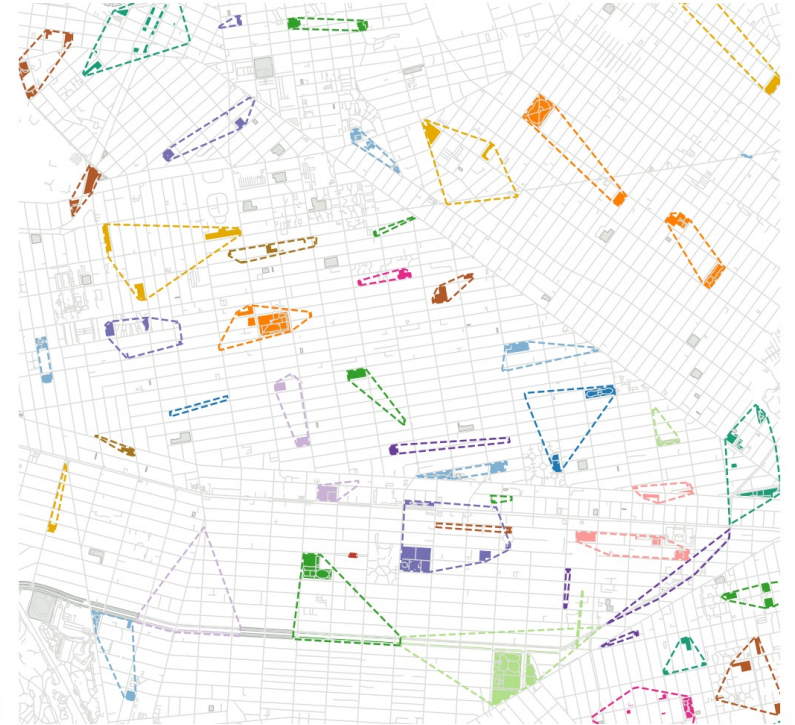
$\gamma = 1$ 336 groups · max 15 parks · median span 395 m

$\gamma = 2$: 340 groups · max 15 parks · median extent 0.48 km



$\gamma = 2$ 340 groups · max 15 parks · median span 391 m

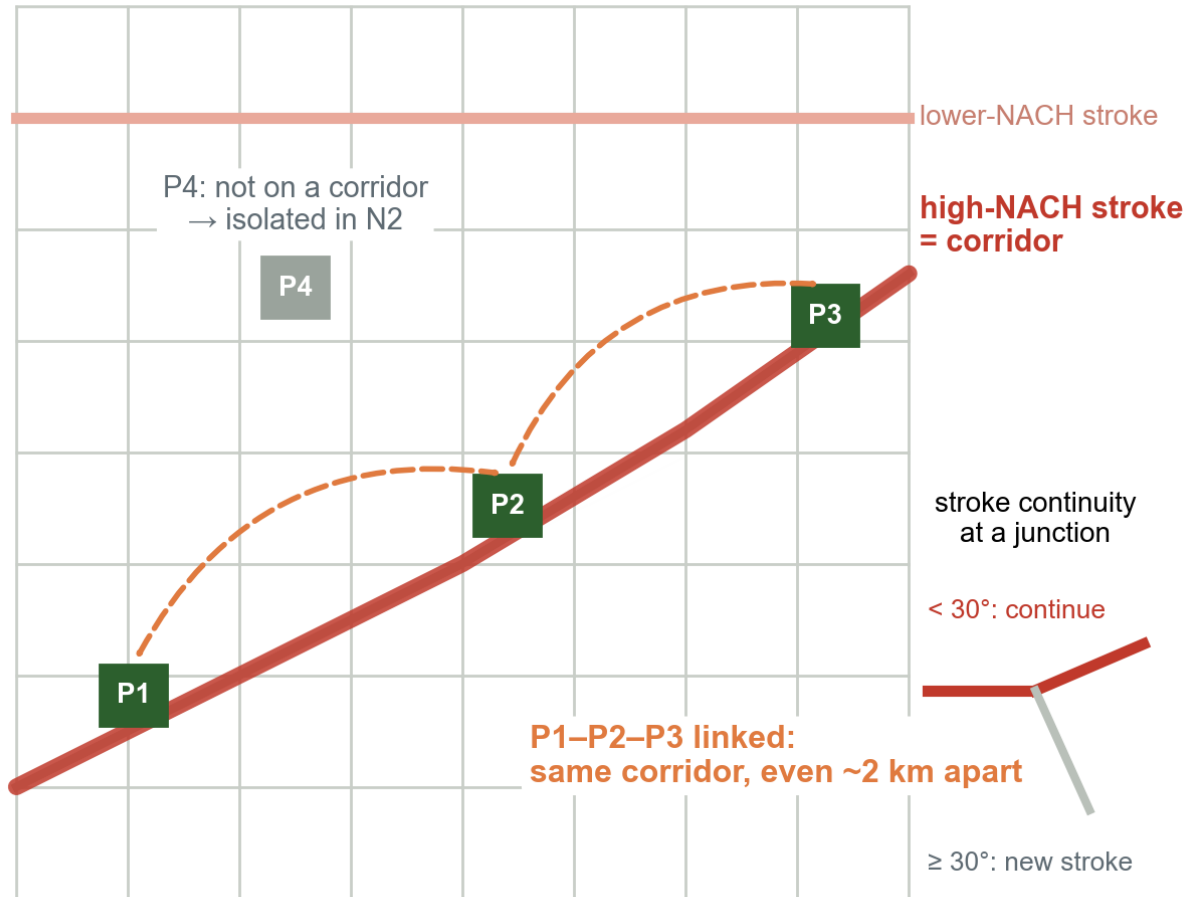
$\gamma = 4$: 345 groups · max 15 parks · median extent 0.47 km



$\gamma = 4$ 345 groups · max 15 parks · median span 370 m

N2 · Shared configurational corridor

Two parks are linked if they sit on the same major movement corridor



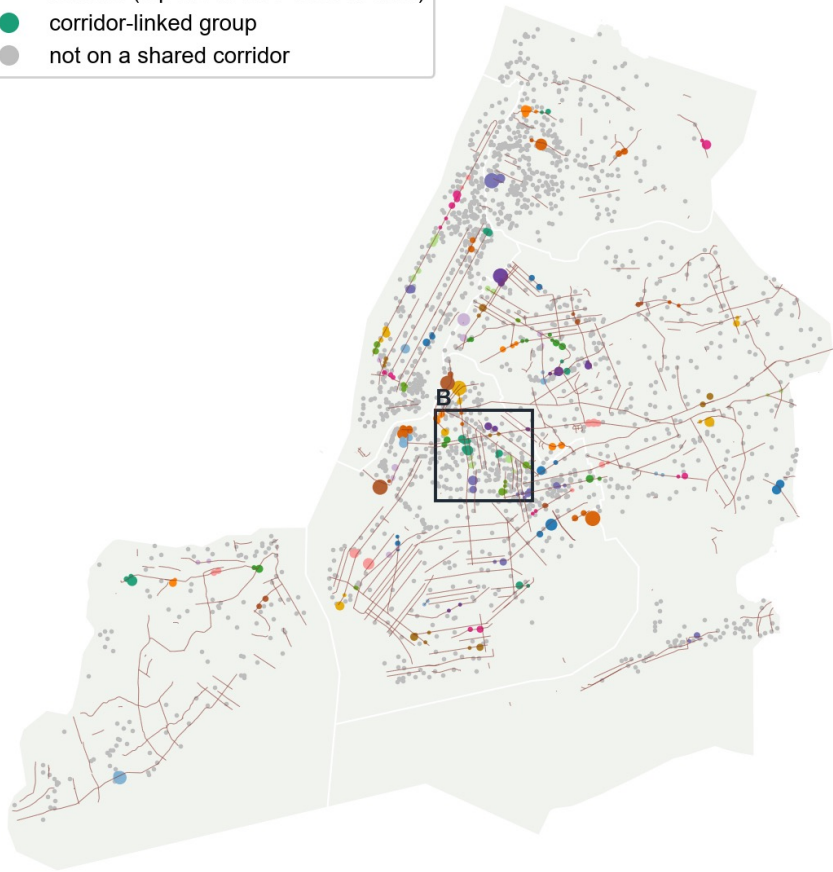
$$\text{NACH}_r = \log(\text{choice}_r + 1) / \log(\text{depth}_r + 3)$$

angular choice and total angular depth from Cityseer
centrality_simplest

- Strokes: chain segments whose direction changes $< 30^\circ$ at each junction (axial-line-like continuity lines)
- Corridor: strokes with the highest mean NACH covering the top p% of network length
- Edge: both parks attach to the same corridor stroke (optional 3.2 km along-corridor limit)
- 18 settings: $r \in \{800, 1600, 3200\} \text{ m} \times p \in \{2, 5, 10\}\% \times \text{limit} \{3.2 \text{ km, none}\}$

N2 · Corridor-linked parks

- corridor (top 5% NACH-1600 strokes)
- corridor-linked group
- not on a shared corridor



B · Fort Greene – Bed-Stuy – Crown Heights



100

corridor groups

253

parks on a shared corridor

371 m

median walking span within a group

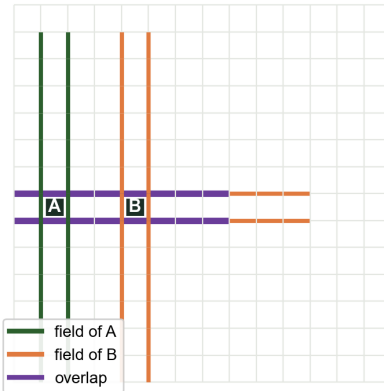
Most parks are off the top corridors — N2 isolation means "not embedded in a major corridor".

N3 · Excess few-turn field overlap

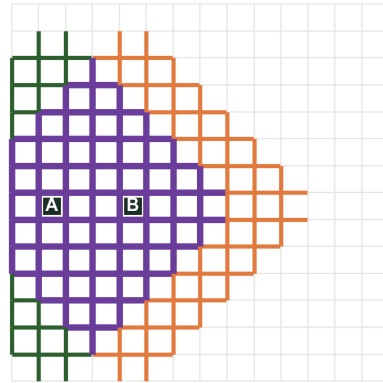
Do two parks share the same few-turn field more than their distance implies?

Near pair (3 blocks): $J_{\text{ang}} = 0.22$, ratio = 0.41 → no edge: proximity explains the overlap

few-turn fields: $J = 0.22$

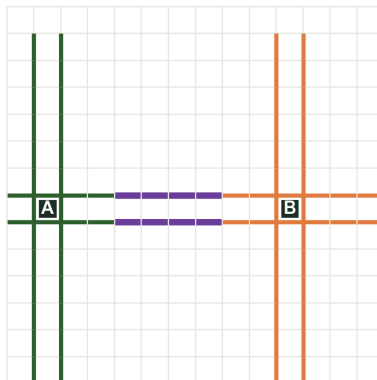


metric fields: $J = 0.52$

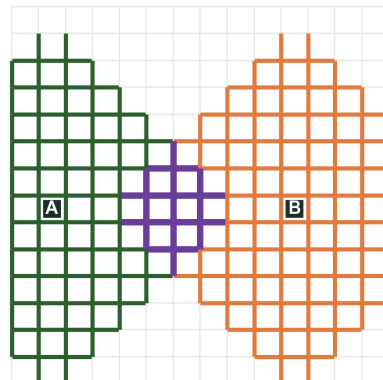


Far pair, same street (9 blocks): $J_{\text{ang}} = 0.10$, ratio = 1.25 → N3 edge ($\tau = 0.1$, $\kappa = 1.0$)

few-turn fields: $J = 0.10$



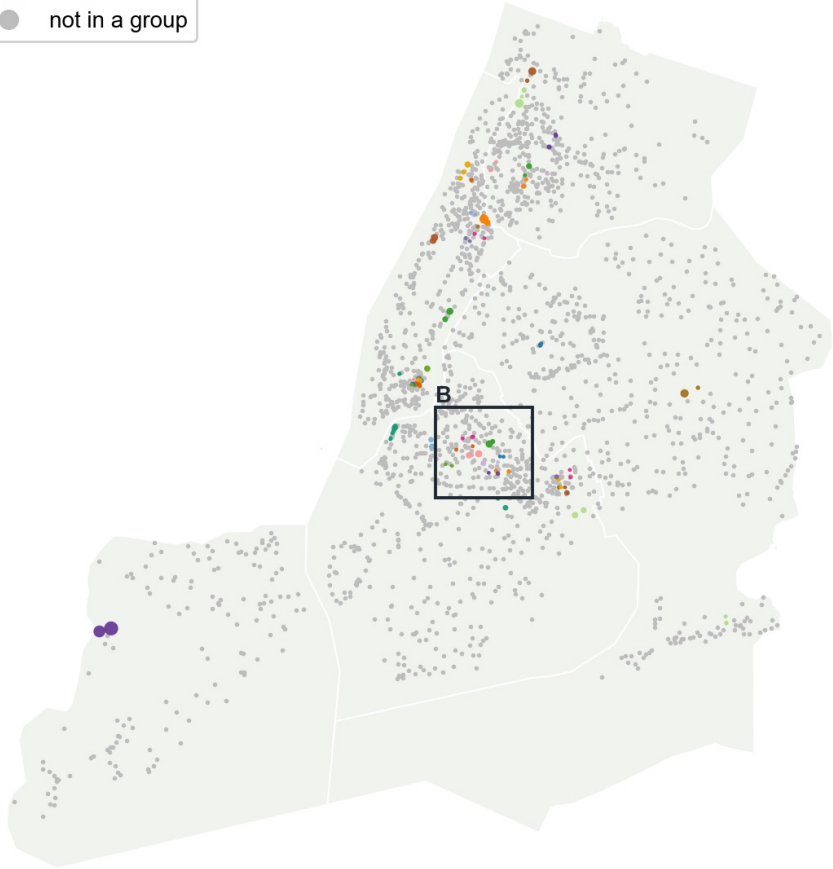
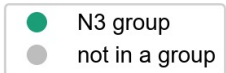
metric fields: $J = 0.08$



- Field = segments reachable with angular cost $\leq \theta c$ and path length $\leq R$
- J = length-weighted Jaccard of two parks' fields; computed for angular and metric fields
- Edge if $J_{\text{ang}} \geq \tau$ and $J_{\text{ang}} / J_{\text{met}} \geq \kappa$
- 16 settings: $\theta c \in \{45, 90\}^\circ \times R \in \{800, 1600\} \text{ m} \times \tau \in \{0.1, 0.2\} \times \kappa \in \{1.0, 1.25\}$

Toy grid: the near pair shares fields only because it is close (ratio 0.41); the far pair on one street keeps sharing its few-turn field (ratio $1.25 > \kappa = 1.0$). Only such pairs become N3 edges, so N3 is sparse.

N3 · A sparse, weak definition



B · Fort Greene – Bed-Stuy – Crown Heights



47

groups (mostly pairs)

8

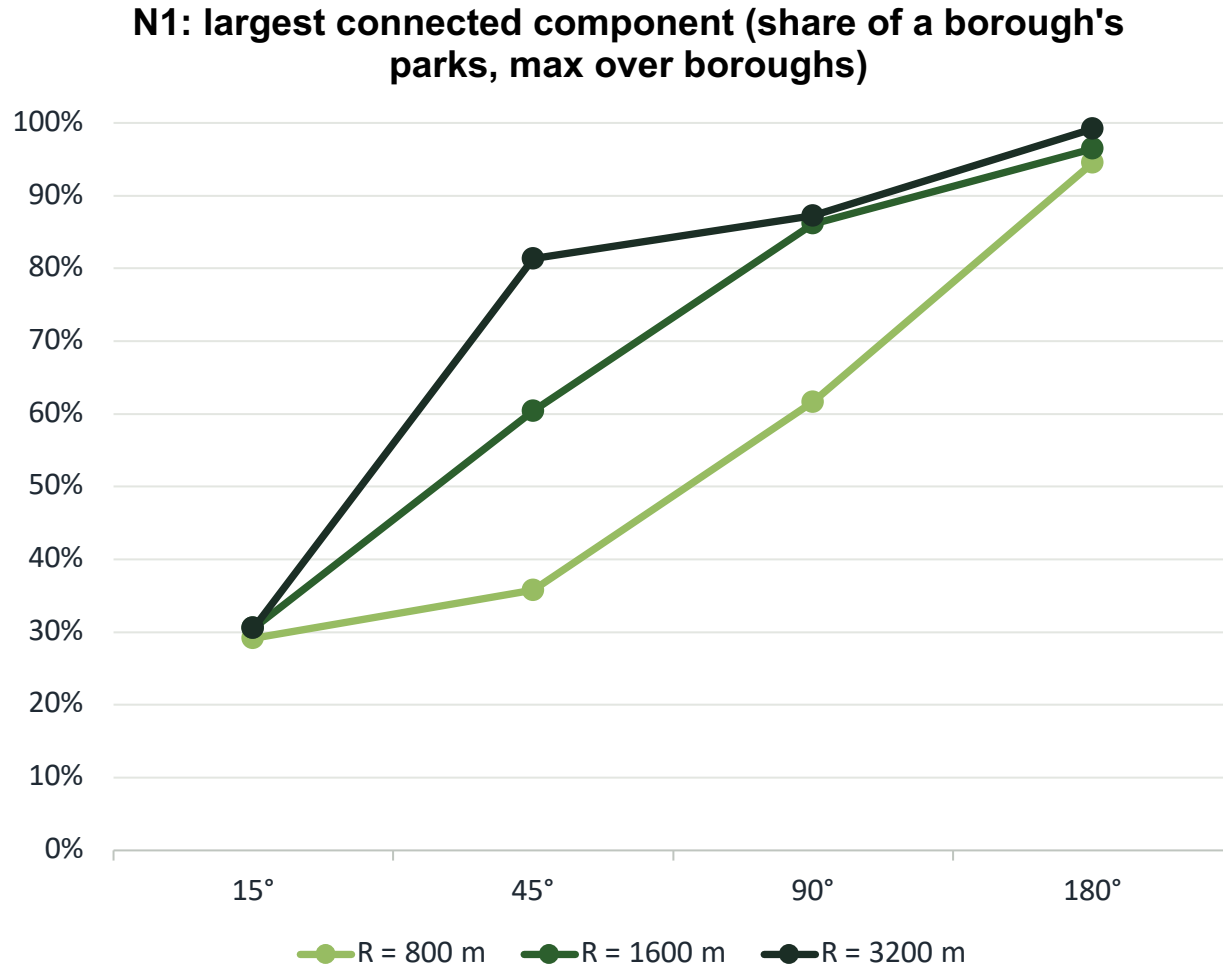
largest group

1,619

parks never in an N3 edge in $\geq 80\%$ of settings

Few-turn fields rarely overlap more than distance predicts: report N3 as weak.

From park graphs to robust groups



Percolation Once $\theta \geq 45^\circ$ at $R \geq 1.6$ km, or one turn is allowed, most of a borough's parks fall into one component — a grid-city finding, but components can no longer separate groups. Queens and Staten Island percolate much later.

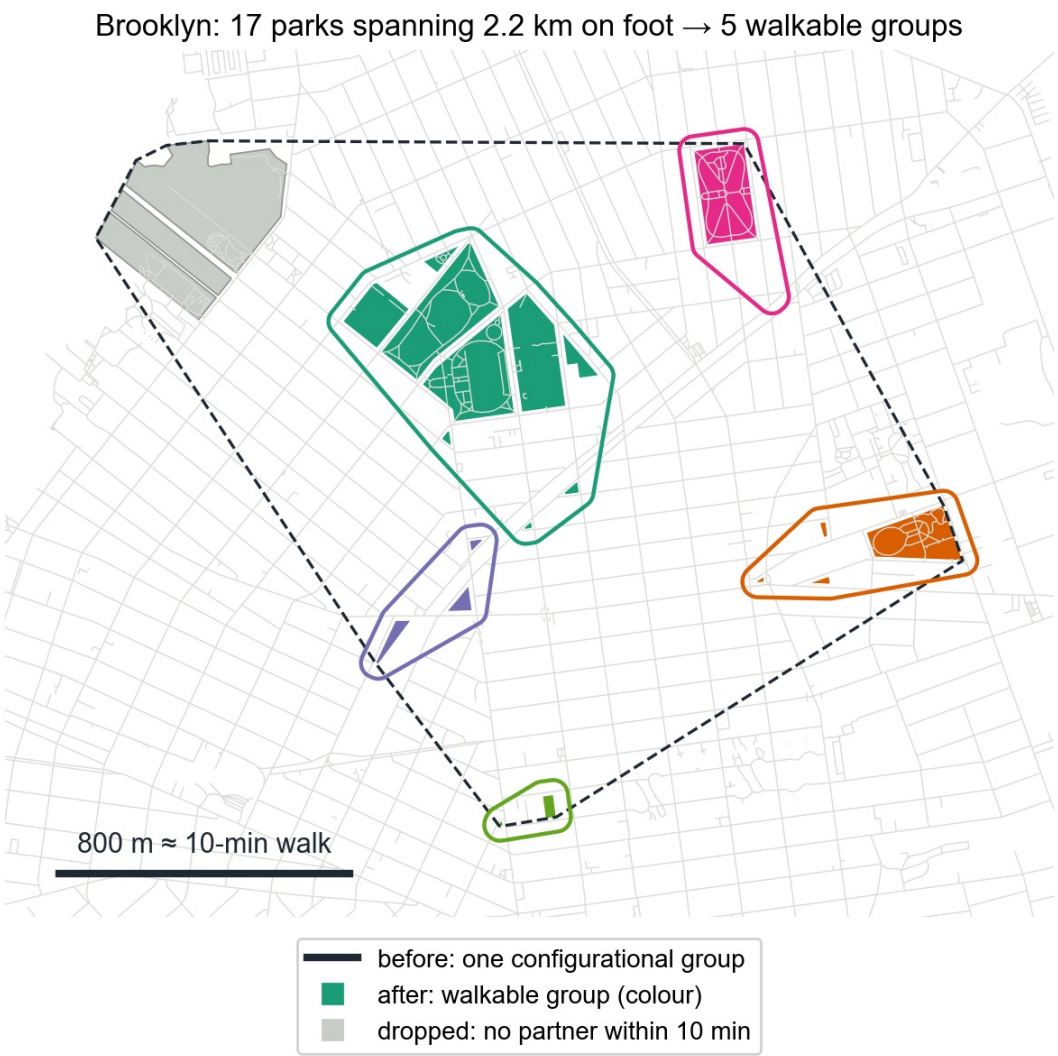
Communities In each setting, Louvain communities find internally dense sets of parks.

Consensus Co-membership share across settings $\rightarrow$ average-linkage clusters (≥ 0.5), then split to a 10-minute walking diameter = robust groups.

Isolation IsolationPersistence = share of settings with no configurational edge; ≥ 0.8 = robust isolated.

Walkable groups: a 10-minute limit

Configuration decides who can be grouped; walking distance caps how far a group can stretch



- Within each consensus group, complete-linkage clustering on shortest walking distance, cut at 800 m
- Every pair of parks in a final group is ≤ 10 minutes apart on foot; parks with no such partner are flagged "beyond walk limit"

	groups	largest group	median walking span
N1 few-turn ($\gamma = 2$)	217 → 340	41 → 15	467 → 391 m
N2 corridor	73 → 100	22 → 7	2,053 → 371 m
N3 field	63 → 47	8 → 8	696 → 414 m

Sensitivity of the walking limit (N1, $\gamma = 2$)

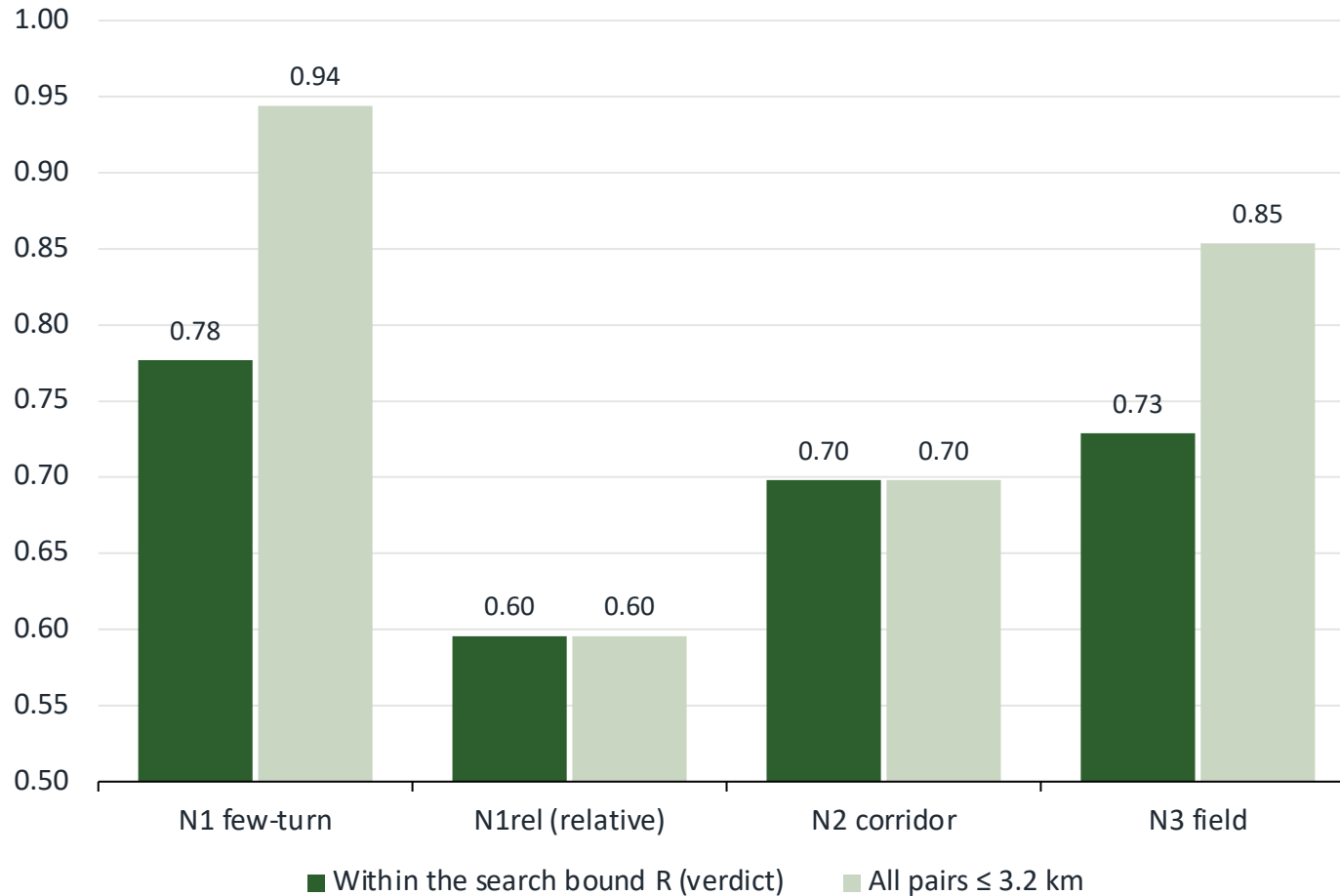
5 min (400 m): 366 groups · 1,013 parks grouped · largest 11

10 min (800 m): 340 groups · 1,178 parks grouped · largest 15

15 min (1200 m): 293 groups · 1,195 parks grouped · largest 22

Is it just proximity?

Can network distance alone predict the configurational edges?

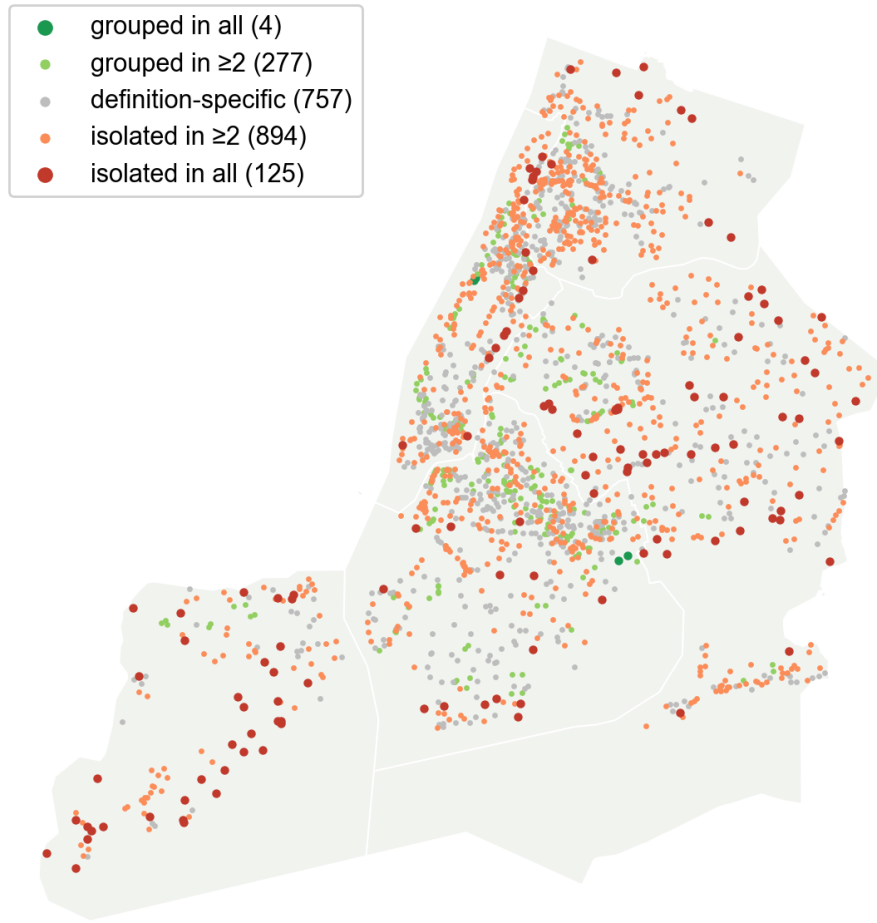


Proximity AUC: probability that a connected pair is closer than an unconnected one; **> 0.85 = proximity-dominated.**

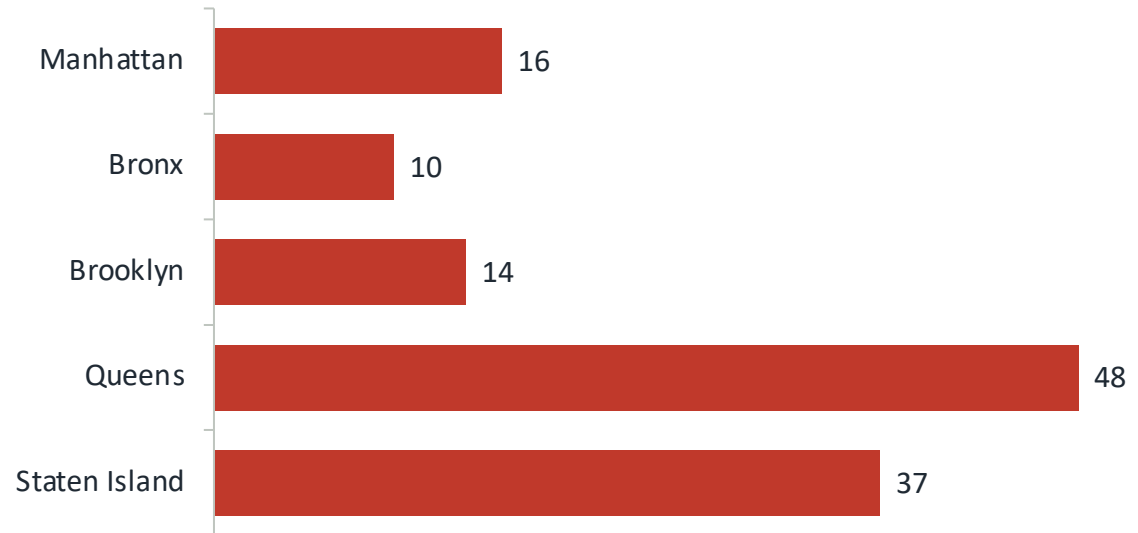
No definition crosses 0.85 once distance is held to its search bound.

Edge overlap with a distance-only graph of equal density: **N1 0.27, N2 0.03, N3 0.00**

Across definitions: who is grouped, who is isolated?



Parks isolated under all three definitions (125)



- The three definitions capture different structure: group agreement between them is near zero ($ARI \approx 0$).
- Only 4 parks are grouped under all three; isolation concentrates in Staten Island and Queens (curvilinear, low-connectivity street patterns) and on expressway-side parkways.

Takeaways & limitations

N1 is the core result

340 walkable few-turn groups ($\gamma = 2$), every pair within a 10-min walk; 132 parks robustly isolated.

Configuration $\neq$ proximity

Within its search bound, distance does not dominate any definition (AUC < 0.85).

Definitions differ

N2 finds corridor chains, N3 is weak; groups barely overlap across definitions.

- Percolation: allowing one turn connects almost all parks in a borough — groups rely on strict settings and community detection
- Turn costs include bends inside segments (one turn $\approx 90\text{--}100^\circ$), so $\theta = 90^\circ$ is a strict one-turn threshold; $\theta \approx 100\text{--}110^\circ$ is worth testing
- OSM completeness and mapping regime (separately drawn sidewalks) affect angular costs; S results differ from C
- N2 computed on C only; large linear parks are split into ~ 400 m pieces; park entrances unknown (boundary attachment)
- Next: validate groups with observed park use / pedestrian counts; test alternative group resolutions per borough